



**Dronabinol Advisory Board
Discussion Summary**

**October 19, 2014
Marriott San Diego Gaslamp Quarter
San Diego, California**

Prepared by Synchrony Healthcare Communications, Inc.

November 4, 2014

Background

INSYS Therapeutics, Inc., held an advisory board on October 19, 2014 to gain insight into current practices regarding Marinol® (dronabinol capsules) prescribing. Nine specialists spanning anesthesiology, oncology, pain management, and physical medicine and rehabilitation discussed their experiences and opinions regarding using dronabinol to treat various problems.

Meeting Objectives

Specifically, the objectives of the advisory board were to gain understanding of the:

- Conditions for which the advisors currently prescribe Marinol
- Factors that may predict individual responses
- Weakness of the current formulation
- Important features for a new formulation

Advisory Board Participants

Ms. Heather Alfonso, APRN
Pain Management
Comprehensive Pain and Headache
Treatment Center
Derby, CT

Jeffrey Gudin, MD
Pain Management
Pain Management Center at Englewood
Hospital and Medical Center
Englewood, NJ

Howard Cohen, MD
Pain Management, Addiction, Psychology
P.R.I.D.E.
Dallas, TX

Brian Kamuda MS-ANP
Oncology
The Block Center
Skokie, IL

Miguel Dominguez, MD
Pain Management
American Pain Institute
Tustin, CA

Ravi Lakkaraju, MD
Physical Medicine and Rehabilitation
Poway Spine and Pain
Poway, CA

Gerald Edelman, MD, PhD
Hematology/Oncology
Medical and Surgical Clinic of Irving
Irvine, TX

Joseph Shurman, MD
Anesthesiology
Scripps Memorial Hospital
La Jolla, CA

Steve Fanto, MD
Physical Medicine and Rehabilitation
Steve Fanto MD PC
Scottsdale, AZ

Thomas Witten, MD
Pain Management
Westmoreland Pain Management Center
Greensburg, PA

INSYS Therapeutics

Michael Babich, MBA
President and Chief Executive Officer

Mark Hein
Senior Marketing Director

Christina Cognata Smith, PharmD, MBA
Medical Director

Jason Kimball
Marketing Director

Synchrony Healthcare Communications

Jonathan Kamien, PhD
Group Medical Lead

Vince Mevoli
Vice President, Client Services

Meeting Format

This meeting was an informal, roundtable discussion led by Michael Babich to draw out the experiences and opinions of the advisors. Mr. Babich began the meeting by welcoming the participants and indicating to the participants that their opinions are needed to inform the marketing plan for INSYS' Dronabinol Oral Solution. He briefly communicated some aspects of the dronabinol market and INSYS' product, including that 80% of Marinol is written for AIDS patients, and that the dronabinol oral solution has been shown to be bioequivalent to Marinol, with a 5-minute onset of action and low variability. The market currently has approximately \$151 million in sales, with 267 million prescriptions and is growing. Launch of INSYS' product is planned for 2015.

Key Learnings and Insights

Conditions for which the advisors currently prescribe dronabinol

- Cancer
 - Nausea (chemotherapy-induced nausea and vomiting)
 - Anorexia/appetite stimulation (30% to 40% of patients have low appetite, or weight loss)
 - Quality of life enhancement (pleasure and joy)
 - Dr. Edelman, in particular, said that dronabinol is underutilized in the cancer population to enhance life enjoyment, which he believes is one of the primary patient needs
 - He uses 5 mg in the elderly; 10 mg in younger or patients experienced with marijuana
 - He assesses for dysphoria whenever prescribing dronabinol, which he said is an anxiety reaction related to paranoid thinking. In cases of dysphoria, he prescribes Xanax to be taken 20 to 30 minutes before dronabinol
- HIV with persistent neuropathy
 - Prescribed for neuropathic pain
 - Recommended to provide in 'a cocktail' along with cannabidiol
- Multiple sclerosis
 - start at lower doses

- Prescribed for neuropathic pain and spasticity
- Complex Regional Pain Syndrome (CRPS)
 - Unexplained weight loss
- Posttraumatic Stress Disorder and chronic neuropathic pain
 - One of the physicians uses this in conjunction with the Wounded Warrior Program

Factors that may predict individual responses

- The advisors agreed that response to Marinol is very variable
 - Marijuana smokers tend to respond quite well
 - A positive history for marijuana use predicts lack of dysphoria
 - Therapeutic response to Marinol can be delayed; typically about 2 weeks from initiation
 - A poor response to Marinol is sometimes associated with functional interference

Weaknesses of the current formulation

- The major weakness of Marinol identified by the advisors is its variability in patient response. This variability exists across patients as well as within patients, with the latter being evidenced by patients reporting that their response is unpredictable, from dose to dose. The advisors believed this to be related to variability in absorption.

Important features for a new formulation

- Insurance authorization for Marinol prescriptions was perceived to be a large barrier
- Studies in advanced illnesses, ie, 'grey zone patients', was recommended
- Quality of life outcome data would be advantageous
- Given the lipophilicity of dronabinol, both fed and fasted pharmacokinetic studies should be conducted
- Patients taking dronabinol for its anti-nausea effects desire a rapid onset and short duration of action

The advisors recommended partnering with Dr. Mark A. Ware of McGill University (<http://www.mcgill.ca/centreformeded/aboutus/whoweare/members/warem>) as an expert in cannabinoids for pain management.